Antenna Radiation Characteristics

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Lecture Objective

By the end of this lecture, you will be able to:

- Evaluate important characteristics of the radiation pattern of a given antenna,
- Understand contributions to the efficiency of an antenna.

Radiation Pattern

- The antenna pattern is a plot of the normalized radiation intensity as a function of direction.
- Most often, we plot the normalized radiation intensity in dB, i.e,

$$10\log_{10}\left(F\left(\theta,\phi\right)\right)$$

Visualization of Radiation Pattern (1 of 2)

- One way to visualize the radiation pattern of an antenna is to plot it in 3-dimensions.

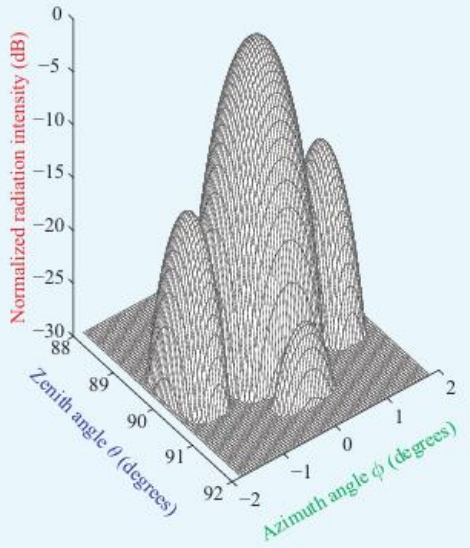


Figure 9-9 Three-dimensional pattern of a narrow-beam antenna.

Visualization of Radiation Pattern (2 of 2)

- Another way to visualize the radiation pattern of an antenna is to plot it in the principle planes in 2-dimensions in either polar or rectangular form.

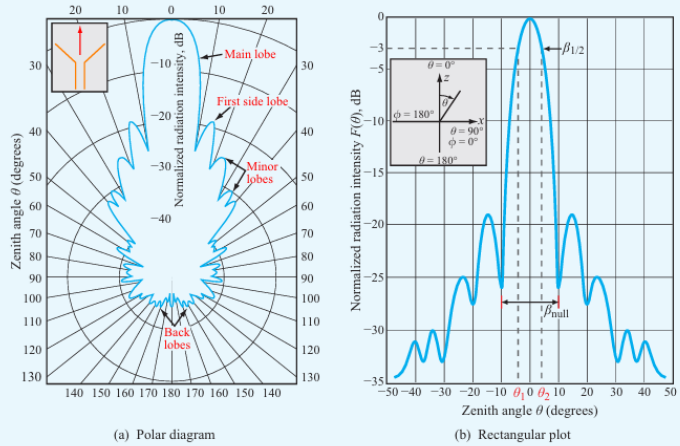
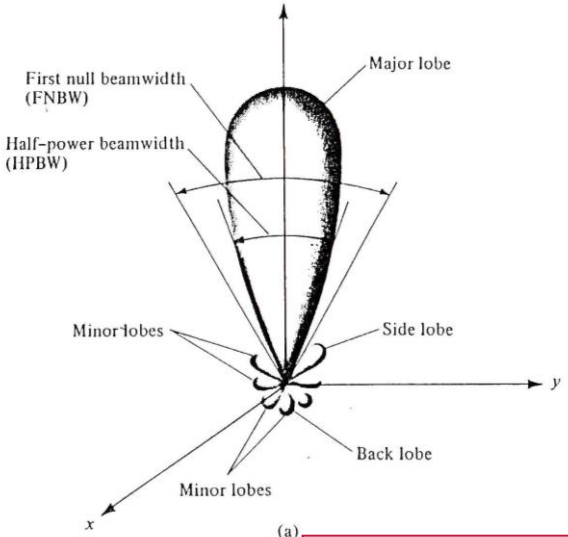


Figure 9-10 Representative plots of the normalized radiation pattern of a microwave antenna in (a) polar form and (b) rectangular form.

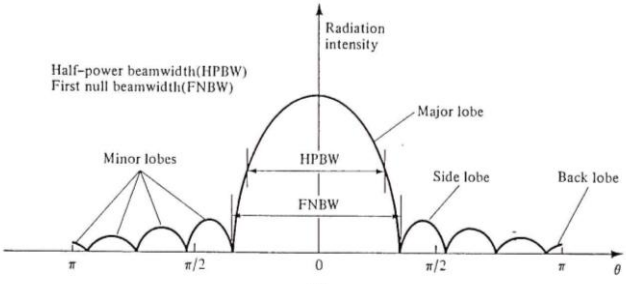
Radiation Pattern Lobes (1 of 2)

- A *radiation lobe* is “a portion of the radiation pattern bounded by regions of relatively weak radiation intensity.”
- The *major lobe* (*main beam*) is “the radiation lobe containing the direction of maximum radiation.”
- In addition, the pattern exhibits side and back lobes which represent (usually) undesirable wasted energy (for transmitting antenna) or potential interference (for receiving antenna).

Radiation Pattern Lobes (2 of 2)



(a)



(b)

Antenna Theory: Analysis and Design, 2nd Edition, Constantine A. Balanis

Beamwidth

- The term ***beamwidth*** is used to describe the angle between two directions in which the radiation intensity satisfies a specified criterion.
- Usually the term “beamwidth” by itself is synonymous with “half-power beamwidth.”
- The ***half-power beamwidth*** (HPBW) is “in a plane containing the direction of maximum of a beam, the angle between the two directions in which the radiation intensity is one-half the maximum value of the beam.”
- The resolution capability of an antenna to distinguish two sources is about equal to the $HPBW$.
- The ***first null beamwidth*** (FNBW) is in a plane containing the direction of the maximum of the beam the angle between the two directions in which the radiation intensity is minimum.

Beam Solid Angle

- ***Beam (pattern) solid angle*** (Ω_p) is the solid angle over which all the radiated power from the antenna would exist if its radiation intensity were constant for all directions within Ω_p and 0 in all other directions.
- It can be evaluated as

$$\Omega_p = \oiint_{\Omega} F(\theta, \phi) d\Omega$$

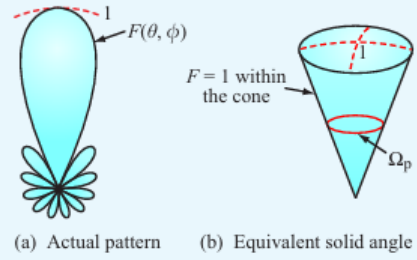


Figure 9-11 The pattern solid angle Ω_p defines an equivalent cone over which all the radiation of the actual antenna is concentrated with uniform intensity equal to the maximum of the actual pattern.

Question 1

- What is the beam solid angle for the Hertzian dipole? Recall that $F(\theta, \phi) = \sin^2 \theta$.

Total Radiated Power

- The total power radiated by the antenna can be computed by adding up the radiation intensity (power per unit solid angle) over all directions (all 4π steradians of solid angle of a sphere):

$$P_{\text{rad}} = \oiint_{\Omega} U(\theta, \phi) d\Omega = U_{max} \oiint_{\Omega} F(\theta, \phi) d\Omega$$

$$= U_{max} \Omega_p$$

Directivity (1 of 2)

- ***Directivity*** is "the ratio of the radiation intensity in a given direction from the antenna to the radiation intensity averaged over all directions. The average radiation intensity is equal to the total power radiated by the antenna divided by 4π."

$$D(\theta,\varphi) = \frac{U(\theta,\varphi)}{\left(P_{\mathrm{rad}}/4\pi\right)}$$

- The term *directivity* is often used to mean the *directivity in the direction of maximum radiation*

$$D = \frac{4\pi U_{max}}{P_{rad}} = \frac{4\pi}{\Omega_p}$$

Directivity (2 of 2)

- Directivity is a figure-of-merit that describes the directional properties of the antenna.
- Note that the directivity of an isotropic element is $D = 1$.
- For all other antennas $D > 1$ in the direction of maximum radiation but less for other directions.

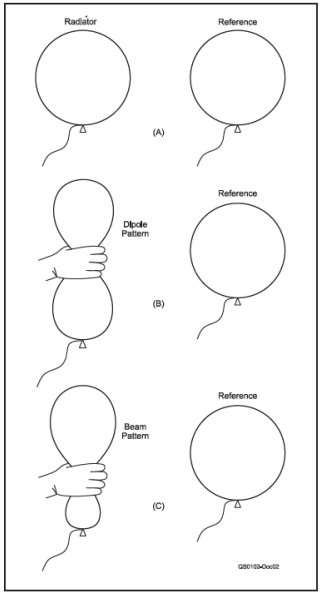


Figure 1.10 — Demonstrating antenna pattern gain with balloons. Take a balloon, blow it up so that it is roughly spherical in shape and declare that it is a radiation pattern from an isotropic radiator. Next, blow up another balloon to the same size and shape — this will be the "reference" antenna (A). Then, squeeze the first balloon in the middle to form a sort of figure-8 shape and a dipole pattern and compare the maximum size to that of the reference "antenna" (B). The dipole can be seen to have some "gain" over the reference isotropic. Next, squeeze the lower end of the dipole balloon to create a shape with more volume toward the top, demonstrating the sort of pattern a beam antenna creates (C).

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Question 2

- What is the directivity of the Hertzian dipole? (Hint: Use the value of beam solid angle that is the correct answer for Question 1.)

Directivity

- Directivity is a dimensionless quantity and is often expressed in dB.

$$D\ (\mathrm{dB}) = 10\log_{10}(D\ (\mathrm{dimensionless}))$$

- When expressed in dB, we often use the designation dBi (dB relative to isotropic) for directivity.

Directivity and Beam Solid Angle

- The beam solid angle of a directional radiation pattern can be estimated as the product of the HPBWs in the two principal planes:

$$\Omega_p \approx \beta_{xz}\beta_{yz}$$

$$D = \frac{4\pi}{\Omega_p} \approx \frac{4\pi}{\beta_{xz}\beta_{yz}}$$

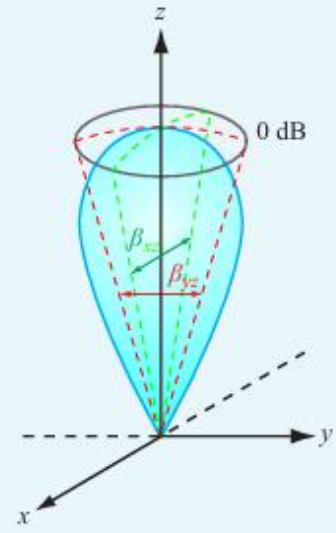


Figure 9-12 The solid angle of a unidirectional radiation pattern is approximately equal to the product of the half-power beamwidths in the two principal planes; that is, $\Omega_p \approx \beta_{xz}\beta_{yz}$.

Gain (1 of 2)

- ***Gain*** is "the ratio of the radiation intensity, in a given direction, to the radiation intensity that would be obtained if the power accepted by the antenna (transmitter power) were radiated isotropically."

$$G(\theta,\varphi) = \frac{U(\theta,\varphi)}{(P_t/4\pi)} = \frac{4\pi\ U(\theta,\varphi)}{P_t}$$

- The term *gain* is often used to mean the gain in the direction of maximum radiation

$$G = \frac{4\pi U_{\max}}{P_t}$$

Gain (2 of 2)

- Gain is a dimensionless quantity and is often expressed in dB.

$$G\ (\mathrm{dB}) = 10\log_{10}(G\ (\mathrm{dimensionless}))$$

- When expressed in dB, we often use the designation dBi (dB relative to isotropic) for gain.

Antenna Efficiency (1 of 2)

- The total power radiated by the antenna is related to the input power as

$$P_{\mathrm{rad}} = \xi P_{\mathrm{t}}$$

radiation efficiency

Antenna Efficiency (2 of 2)

- Thus, the relationship between gain and directivity is

$$G = \xi D$$

- Since $0 \leq \xi \leq 1$ we must have

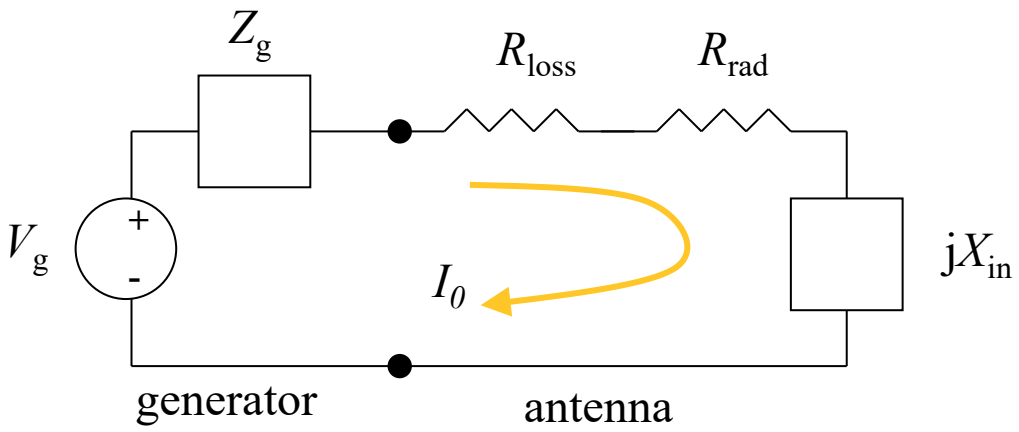
$$G(\theta, \varphi) \leq D(\theta, \varphi)$$

Matched and Realized Gain

- The term ***matched gain*** is often used in practice to mean gain as described in the current IEEE standard.
- The term ***realized gain*** is often used in practice to mean a "gain" which takes into account the mismatch at the antenna terminals *as well as the polarization mismatch*.
- Realized gain is what is actually measured in the antenna range.

Radiation Resistance

- From the point-of-view of the transmitter, the antenna is a load with a certain input impedance.



Radiation Resistance & Radiation Efficiency

$$Z_{\text{in}} = R_{\text{in}} + jX_{\text{in}} = R_{\text{rad}} + R_{\text{loss}} + jX_{\text{in}}$$

$$P_t = \frac{1}{2} I_0{}^2 R_{\text{in}} = \underbrace{\frac{1}{2} I_0{}^2 R_{\text{rad}}}_{P_{\text{rad}}} + \underbrace{\frac{1}{2} I_0{}^2 R_{\text{loss}}}_{P_{\text{loss}}} \Rightarrow R_{\text{rad}} = \frac{2P_{\text{rad}}}{I_0{}^2}$$

$$\xi = \frac{P_{\text{rad}}}{P_{\text{t}}} = \frac{P_{\text{rad}}}{P_{\text{rad}} + P_{\text{loss}}} = \frac{R_{\text{rad}}}{R_{\text{rad}} + R_{\text{loss}}}$$

Radiation Resistance & Radiation Efficiency of Hertzian Dipole

$$P_{\text{rad}} = 40\,\pi^2 I_0^2 \left(\frac{l}{\lambda}\right)^2 \Rightarrow R_{\text{rad}} = \frac{2P_{\text{rad}}}{{I_0}^2} = 80\,\pi^2 \left(\frac{l}{\lambda}\right)^2$$

Assume antenna comprises a wire of circular cross section of radius a. Because of skin effect, the cross-sectional area over which current flows is the perimeter of the wire times the skin depth.

$$R_{loss} = \frac{l}{\sigma_c A} = \frac{l}{2\pi a \delta_s \sigma_c} = \frac{l}{2\pi a}\sqrt{\frac{\pi f \mu_c}{\sigma_c}}$$

$$A = 2\pi a\,\delta_s$$

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